

20260330 & 20260401EB_MN

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2026年3月30日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates (WT11, P8, 5wells)
keep some iPSCs

2026年4月1日 星期三

T2 -> T3
Start differentiation for JiPSC1000, 1066, 1108 (T2)

2026年4月3日 星期五

WT11: T3 -> T4 (~14 mL for over-weekend)
JiPSC1000, 1066, 1108, 5 wells in 10-cm dish: T2 -> T3

2026年4月5日 星期日

JiPSC1000, 1066, 1108: T3 -> T4

2026年4月6日 星期一

WT11: T4 -> T5

2026年4月7日 星期二

JiPSC1000, 1066, 1108: add 4mL T4

2026年4月8日 星期三

WT11: T5 -> T6
JiPSC1000, 1066, 1108: T4 -> T5

2026年4月10日 星期五

WT11: T6 -> T6 (~14mL for over-weekend)
JiPSC1000, 1066, 1108: T5 -> T6

2026年4月13日 星期一

T6 -> T7

Dissociation

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 4mL 0.25% trypsin (prepare 8mL 0.25% trypsin with 220U DNaseI). **It's new trypsin!!!!**
3. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
4. pipette*10 at 5min, 10min (for JiPSC1000, 1066, 1108: stop after pipetting at 10min) , and 15min (for WT11, but the EBs are not quite visible at 10min. Perhaps I should stop at 10min since I got low yield)
5. add 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times

6. add 8mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.
8. centrifuge at 200xg, 3.5min (Try to see if I can avoid having some cell clumps after resuspension, but I still have cell clumps after pipetting. It might because of excessive mechanical stress....)

*Note: newly opened trypsin is very easy to over digest cells....

cell info				^
	A	B	C	
1	cell	passage	well #	
2	WT11	P8	5	
3	JIPSC1000	P3+4	5	
4	JIPSC1066	P6+4	5	
5	JIPSC1108	P6+4	5	

Table1			^
	A	B	
1		volume (mL)	
2	trypsin	4	
3	HI-FBS/CTWM	8	

Table3											/
	A	B	C	D	E	F	G	H	I	J	
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	1000	4.26	99	7	29.82	A	96	60	36	507	
3						B	24	250	16	939	
4						C,D	6	1972	12	the rest	

Well2							^
	1	2	3	4	5	6	
A	WT11						
B	iPSC1000						
C	iPSC1066						
D	iPSC1108						

Well3							^
	1	2	3	4	5	6	
A	185 0.5uL	185 2uL	185 5uL				
B	187 0.5uL	187 2uL	187 5uL				
C							
D							

2026年4月14日 星期二

Change T7 -> T7 if added virus

iPSC1000 and iPSC1108 grew very bad

2026年4月15日 星期三

1/2 T7-> 1/2 T7 if not added virus

2026年4月16日 星期四

T7 -> T8

2026年4月17日 星期五

1/2 T8 -> 1/2 T8

2026年4月19日 星期日

1/2 T8 -> 1/2 T8

2026年4月21日 星期二

1/2 T8 -> 1/2 T8

2026年4月22日 星期三

Treat IF cells

Well1

	1	2	3	4	5	6
A	2.5mM gluc 1h	0mM gluc 1h	2.5mM gluc 3h	0mM gluc 3h	2.5mM gluc 6h	0mM gluc 6h
B						
C						
D						

Table2

	A	B	C	D
1		treat	fix	done?
2	2.5mM gluc 1h	17:00	18:00	
3	0mM gluc 1h	17:00	18:00	
4	2.5mM gluc 3h	15:00	18:00	
5	0mM gluc 3h	15:00	18:00	
6	2.5mM gluc 6h	12:00	18:00	v
7	0mM gluc 6h	12:00	18:00	v

prepare media with different conc. of glucose

	A	B	C	D	E	F
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (mL)	added volume (uL)	media volume
2	1100	2.5	440	12	27.3	11972.7
3		0		12		

Check virus test wells

Cells looked much better! I should trash the virus I made.

Table4					
	A	B	C	D	E
1	virus volume (uL)	185	note for 185	187	
2	0.5	many cells are mCherry+. With 30%, 300ms, signal is ~120-200. Mostly ~120-150. Have some cytosolic puncta.		With 30%, 300ms, 488, the signal is around 120. Very few cells reach 150. Signal was very bad in general and still a lot of cytosolic puncta.	
3	2	Signal is much brighter (150+). Some cells are 170+. Brighter and more cytosolic puncta. I think this MOI is good.	I took a movie, but I accidentally set 561 200ms, which is too short. (and use 20% 488) But can see the activation.	With 30%, 300ms, 488, the signal is around 120-130. A few cells can reach 150. Still have a lot of cytosolic puncta.	
4	5	Signal is even brighter, but not quite different from 2uL. Cells look okay, but they seems to be healthier with 2uL.		A lot of cytosolic puncta. However, the signal was very low ~120-130 with 20%, 300ms, 488. With 30%, 300ms, 488, the signal is better ~140, a few cells have ~180.	I don't think the signal is good enough to measure. Maybe try to clone it into our backbone?

IF

- 1. 4%PFA, 15min at RT
- 2. Wash with PBS once
- 3. 0.2%Triton for 10min at RT
- 4. Blocking from 18:50-20:40
- 5. Treat 1st Ab

Table6														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	condition	Ab#	antibody	species	dilution	volume/well	well#	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	33	TDP-43	Rb	500	162.5	24	3900	7.8	488	B	1000	4000	4

2026年4月23日 星期四

- 1. PBS wash 10min*3
- 2. 2nd Ab (C: 1:1000) for 1h at RT
- 3. PBS wash *3
- 4. Check cells: some wells were partially detached.
- 5. 1ug/ml DAPI 10min
- 6. PBS wash 5min *3

2026年4月24日 星期五

Imaging IF

IF_O.C. , 60X, 600x900, 16-bit (Nikon Ti2)					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	405	16	10	100 ms	
3	488	16	30	300 ms	
4	DIC	12			

fileID_treatment_IF							
	A	B	C	D	E	F	G
1	image	treatment	treat_time	488	cell	pyruvate	
2	MN_488-TDP43_405-DAPI_60x	2.5mM gluc	1h	TDP43	WT11	no	
3	MN_488-TDP43_405-DAPI_60x_0001	0mM gluc	1h	TDP43	WT11	no	
4	MN_488-TDP43_405-DAPI_60x_0002	2.5mM gluc	3h	TDP43	WT11	no	
5	MN_488-TDP43_405-DAPI_60x_0003	0mM gluc	3h	TDP43	WT11	no	
6	MN_488-TDP43_405-DAPI_60x_0004	2.5mM gluc	6h	TDP43	WT11	no	
7	MN_488-TDP43_405-DAPI_60x_0005	0mM gluc	6h	TDP43	WT11	no	
8	MN_488-TDP43_405-DAPI_60x_0006	2.5mM gluc	1h	TDP43	iPSC1066	no	
9	MN_488-TDP43_405-DAPI_60x_0007	0mM gluc	1h	TDP43	iPSC1066	no	
10	MN_488-TDP43_405-DAPI_60x_0008	2.5mM gluc	3h	TDP43	iPSC1066	no	
11	MN_488-TDP43_405-DAPI_60x_0009	0mM gluc	3h	TDP43	iPSC1066	no	a part of the wells detached
12	MN_488-TDP43_405-DAPI_60x_0010	2.5mM gluc	6h	TDP43	iPSC1066	no	
13	MN_488-TDP43_405-DAPI_60x_0011	0mM gluc	6h	TDP43	iPSC1066	no	
14	MN_488-TDP43_405-DAPI_60x_0012	0mM gluc	6h	TDP43	iPSC1000	no	
15	MN_488-TDP43_405-DAPI_60x_0013	2.5mM gluc	6h	TDP43	iPSC1000	no	
16	MN_488-TDP43_405-DAPI_60x_0014	2.5mM gluc	6h	TDP43	iPSC1108	no	
17	MN_488-TDP43_405-DAPI_60x_0015	0mM gluc	6h	TDP43	iPSC1108	no	didn't have many cells alive. very sparse

I found it's hard to define the cytoplasmic part (especially when I am imaging iPSC1066)

--> still take some prilimenary photos (~10 fields/condition for iPSC1066)

--> stain TUBB (mouse) to get cytosolic signal for better segmentation

Table7														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	condition	Ab#	antibody	specie s	dilution	volume/well	well#	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	c-28	Purified anti-Tubulin β 3 (TUBB3) Antibody [TUJ1]	m	200	162.5	24	3900	19.5	488	B	1000	4000	4